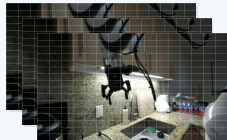
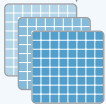


1 Observe



frames $t-2, t-1, t$

frozen DINO encoder E



patch grids $\mathbf{Z}_{-2:0}$

$16 \times 16 \text{ tokens} \times C$

2 Condition on actions

all horizons $k=1 \dots K$ in parallel

Memory encoder

space-time self-attention
 $3 \times 256 \text{ tokens} \cdot L_e \text{ layers}$

Query decoder

self- & cross-attention
 $\text{AdaLN}(\mathbf{c}_k) \cdot L_d \text{ layers}$

Causal prefix GRU

$\mathbf{c}_k$ sees only $a_{<k}$

a_{-2}

a_{-1}

a_0

a_1

$\dots$

a_{k-1}

past

future action prefix

past actions

$\mathbf{c}_k$

$\mathbf{h}_k$

3 Move what was seen, then correct

(a) Local transport

$w \times w$ window, S frames
 $\mathbf{T}_k = \pi_k \mathbf{Z}$



(b) Gate

$g_k = \sigma(\mathbf{w}^\top \mathbf{h}_k)$
move or stay, per patch



(c) Correction $\mathbf{r}_k = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_k$

content transport cannot explain

$$\hat{\mathbf{Z}}_k = (1 - g_k) \mathbf{Z}_0 + g_k \mathbf{T}_k + \mathbf{r}_k$$

$\hat{\mathbf{Z}}_1, \dots, \hat{\mathbf{Z}}_K$



+

$\mathbf{Z}_0$

measured features are moved and kept — every horizon is anchored to observation, never to its own forecast